Cohomology in AC/DC RLC Circuits and Measurement

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Abstract

We present a cohomological viewpoint on lumped AC/DC RLC circuits, organized around a two-layer separation: topology determines which state spaces and pairings exist, while geometry and constitutive data (via a Hodge/constitutive operator and energy functionals) determine numerical parameters such as R, L, and C. At the lumped level, Kirchhoff constraints are expressed using chains/cochains on a graph, with the first cohomology measuring obstructions to globally defined potentials. At the field-domain level, Alexander duality and Poincaré–Lefschetz duality identify capacitive and inductive degrees of freedom and interpret idealized measurement as evaluation pairings. Periodic time (S^1) organizes AC steady states via Künneth-type decompositions and universal $i\omega$ phase factors.

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1 Introduction

Electrical circuits are often taught as an exercise in solving linear equations, but there is a persistent geometric theme: conservation laws look like exactness, potentials exist when cohomology vanishes, and interconnection is a gluing operation. The goal of this note is to make that theme concrete for lumped AC/DC RLC circuits, in a way that is intentionally pedagogical rather than encyclopedic.

1.1 Motivation and a two-layer viewpoint

Our guiding separation principle is:

- **Topology organizes existence.** (Co)chain complexes, exact sequences, and duality determine *which* degrees of freedom are possible (and their dimensions).
- **Geometry supplies numbers.** A metric and material parameters enter through a Hodge/constitutive operator and energy/dissipation functionals; numerical parameters such as R, L, and C are matrices of the resulting bilinear forms in physically meaningful bases [1, 2].

We emphasize two complementary models:

- a *lumped graph model* (a finite oriented 1-complex) capturing Kirchhoff constraints, terminal behavior, and basic compositional rules; and
- a *field-domain model* $U = S^3 \setminus \mathrm{int}(M)$ capturing the topological degrees of freedom behind capacitance and inductance via Alexander duality and Poincaré–Lefschetz duality.

1.2 Related work (brief)

This project is strongly influenced by the compositional/variational viewpoint of Baez–Fong and collaborators on passive linear networks [3, 4]. Our emphasis differs mainly in that we keep the ambient 3-dimensional topology in view (to explain energy storage and linking phenomena) and we treat time periodicity via a space–time product picture (rather than, say, Laplace transform as a primary language).

For topological electromagnetics and the use of duality/linking in computational settings, see Gross–Kotiuga [5]. For the functional-analytic/discretization perspective where Hodge theory, boundary conditions, and spectral gaps matter (useful for the "almost topological" discussion), see finite element exterior calculus (FEEC) [6].

1.3 Organization and reading guide

Section 2 collects notation and the specific algebraic-topology tools we use. Section 3 develops the lumped graph model and proves the Kirchhoff statements. Section 4 introduces the field-domain model, terminals, and measurement as duality pairings. Section 5 records excision and Mayer–Vietoris and explains how Dirichlet-to-Neumann maps and Schur complements encode "black-box"

behavior under interconnection. Sections 6 and 7 treat the two main organizing mechanisms for RLC behavior beyond Kirchhoff bookkeeping: Alexander duality (capacitive/inductive state spaces) and time topology (AC vs. DC). Section 8 contains worked examples and an optional "almost topological" module.

1.4 Main statements (informal but precise)

The following statements are the spine of the paper; later sections provide proof sketches and examples.

Theorem 1.1 (Kirchhoff laws as boundary/coboundary conditions)**.** *Let $X = (V, E)$ be a finite oriented graph and let $T \subset V$ be a set of* terminal *vertices. Fix a coefficient field R (typically $R = \mathbb{R}$). Currents are 1-chains $i \in C_1(X; R)$ and voltages are 1-cochains $v \in C^1(X; R)$. Then:*

- ***(KCL)*** *Current conservation at internal nodes is the condition*

$$\partial i = 0 \quad \text{in } C_0(X, T; R),$$

  i.e. ∂i is supported on terminals.

- ***(KVL)*** *The existence of a node potential $\phi \in C^0(X; R)$ with $v = d\phi$ is equivalent to*

$$v(\gamma) = 0 \quad \text{for all cycles } \gamma \in Z_1(X; R),$$

  i.e. to $[v] = 0$ in $H^1(X; R)$.

Theorem 1.2 (AC adds time-circle degrees of freedom)**.** *Let Y be a space (or pair) modeling spatial degrees of freedom (e.g. (X, T) or (U, Γ)), and let T be either an interval $[0, \infty]$ (DC/transients) or a circle S^1 (steady periodic forcing). Over a field, the (co)homology of $Y \times T$ decomposes via Künneth as a direct sum of tensor products. In particular, replacing a contractible time axis by a time circle adds a new $H_1(S^1)$ factor, which is the topological shadow of additional AC/phasor degrees of freedom.*

Theorem 1.3 (Capacitor/inductor degrees of freedom via Alexander duality)**.** *Let $M \subset S^3$ be a compact conductor region with field domain $U = S^3 \setminus \mathrm{int}(M)$. Then the degrees of freedom governing electrostatics and magnetostatics are naturally expressed in terms of reduced homology of M and (co)homology of U, related by Alexander duality. With terminals $\Gamma \subset \partial U$ specified, the relevant spaces are relative groups and the associated Poincaré–Lefschetz pairings.*

Theorem 1.4 (Measurement as a duality pairing)**.** *Let U be an oriented compact 3-manifold-with-boundary and let $\Gamma \subset \partial U$ be a union of terminal boundary components. Poincaré–Lefschetz duality supplies perfect evaluation pairings of the form*

$$H^k(U, \Gamma; R) \times H_{3-k}(U, \partial U \setminus \Gamma; R) \longrightarrow R,$$

which can be interpreted as "effort evaluated on flow". After choosing a constitutive Hodge operator $\star_{\mathrm{mat}}$, these pairings induce inner products and hence canonical projections/minimization principles underlying idealized readout.

Theorem 1.5 (Gluing, exact sequences, and Schur complements)**.** *Suppose a device is built by gluing two subdevices along an interface (at the level of graphs or field domains). Then Mayer–Vietoris and related long exact sequences describe how the relevant (co)homology state spaces split or are constrained under gluing. If the device is described by a quadratic energy/dissipation functional, eliminating internal interface degrees of freedom produces a Schur complement, yielding an effective terminal operator (a Dirichlet-to-Neumann map) compatible with interconnection.*

2 Background, notation, and conventions

We assume basic familiarity with singular/cellular (co)homology, long exact sequences, and duality; standard references include [7, 8]. For differential forms and the Hodge-star viewpoint used to connect topology to energy pairings, see [2, 9].

2.1 Notation and coefficients

Unless stated otherwise we work over $R = \mathbb{R}$. In AC steady state we sometimes extend scalars to $\mathbb{C}$ (phasors). We write $\widetilde{H}_\bullet$ and $\widetilde{H}^\bullet$ for reduced (co)homology.

2.2 Chains and cochains on a graph

Throughout, a lumped circuit is modeled as a finite oriented 1-dimensional CW complex (a graph)

$$X = (V, E).$$

Cellular chain groups are $C_k(X; R)$ with boundary $\partial\colon C_1(X; R) \to C_0(X; R)$, and cochains are $C^k(X; R) = \mathrm{Hom}(C_k(X; R), R)$ with coboundary $d\colon C^0(X; R) \to C^1(X; R)$. Evaluation provides the basic pairing $C^1(X; R) \times C_1(X; R) \to R$.

2.3 Relative (co)homology and terminals

Graph terminals. To speak about a *device* rather than a closed circuit, we fix a set of terminal vertices $T \subset V$, viewed as a subcomplex. Relative chain groups $C_\bullet(X, T; R)$ encode "internal conservation" with boundary allowed on terminals; for example KCL at internal nodes is $\partial i = 0$ in $C_0(X, T; R)$. The relative groups $H_\bullet(X, T; R)$ and $H^\bullet(X, T; R)$ carry a natural Kronecker pairing

$$H^k(X, T; R) \times H_k(X, T; R) \to R,$$

which is the clean home for "readouts between terminals".

Field terminals. In the field-domain model, terminals are represented by a union $\Gamma \subset \partial U$ of boundary components where boundary conditions/probes are imposed. The pair (U, Γ) determines relative groups such as $H^\bullet(U, \Gamma; R)$ and $H_\bullet(U, \partial U \setminus \Gamma; R)$, which are dual under Poincaré–Lefschetz duality.

2.4 Exact-sequence toolkit

We use:

- the long exact sequence of a pair (X, A),
- excision (local modifications away from terminals do not change relative groups), and
- Mayer–Vietoris long exact sequences for decompositions $X = A \cup B$,

to formalize how constraints and degrees of freedom behave under interconnection. We also use functoriality: inclusions induce maps on (co)homology, and most constructions below are natural with respect to these maps.

2.5 Dualities used in the paper

Alexander duality (in S^3). For a compact locally contractible $A \subset S^3$ and coefficient ring/field R, Alexander duality provides

$$\widetilde{H}^k(S^3 \setminus A; R) \cong \widetilde{H}_{2-k}(A; R).$$

Poincaré–Lefschetz duality. For an oriented compact 3-manifold-with-boundary U and $\Gamma \subset \partial U$, there are perfect pairings

$$H^k(U, \Gamma; R) \times H_{3-k}(U, \partial U \setminus \Gamma; R) \to R$$

given by evaluation/integration.

2.6 Cup, cap, and slant products (optional)

The most frequently used "product" in this note is evaluation: a cohomology class is measured on a homology class. At the level of relative groups this is the Kronecker pairing

$$H^k(X, A; R) \times H_k(X, A; R) \to R,$$

and more generally it can be expressed using the cap product (and, dually, cup products) [7, 2].

Cup products (where they can matter). On a graph X (a 1-complex), $H^2(X; R) = 0$, so the cup product of two H^1 classes necessarily vanishes. In contrast, for a 3-dimensional field domain U one can have nontrivial products such as

$$H^1(U; R) \times H^1(U; R) \to H^2(U; R),$$

and under Poincaré–Lefschetz duality these products are closely related to intersection/linking-type pairings (heuristically: "multiply two circulations to get a flux class").

Slant products (integration along a factor). For a product $Y \times S^1$, Künneth describes how classes decompose into spatial and temporal factors. The slant product with the fundamental class $[S^1] \in H_1(S^1; R)$ provides a canonical "integration over time" map

$$H^{k+1}(Y \times S^1; R) \to H^k(Y; R),$$

which is the algebraic-topology avatar of taking a period integral / time-averaged quantity in the de Rham picture. We return to this viewpoint in the AC discussion in §7.1.

2.7 Hodge star and constitutive data

Topology alone produces (co)homology groups describing *which* fluxes/charges/circulations are possible. Numerical parameters require geometric/constitutive data, typically encoded by a Hodge/constitutive operator $\star_{\mathrm{mat}}$ and the associated inner product

$$\langle \alpha, \beta \rangle_{\mathrm{mat}} := \int_U \alpha \wedge (\star_{\mathrm{mat}} \beta).$$

2.8 Time topology: $[0, \infty]$ versus S^1

We model time dependence by products with a 1-manifold T:

- *DC/transients:* $T = [0, \infty]$ (contractible, with boundary), and
- *AC steady periodic forcing:* $T = S^1$ (periodic, no boundary).

Künneth decompositions describe how state spaces change under $Y \mapsto Y \times T$. A useful caution is that homology of S^1 only sees the constant class and the winding class; the full Fourier spectrum of a periodic signal enters analytically (as eigenfunctions/characters), not through singular homology alone.

3 The lumped graph model

3.1 Model and dictionary

Let $X = (V, E)$ be a finite oriented graph. We interpret:

- branch currents as 1-chains $i \in C_1(X; R)$,
- node potentials as 0-cochains $\phi \in C^0(X; R)$, and
- branch voltages as 1-cochains $v \in C^1(X; R)$.

If terminals $T \subset V$ are specified, then $C_\bullet(X, T; R)$ encodes "internal" conservation with boundary allowed on terminals.

Circuit notion	Topological/algebraic notion
Branch current i	1-chain $i \in C_1(X; R)$
Node current injection	$\partial i \in C_0(X; R)$ (supported on T)
Node potential ϕ	0-cochain $\phi \in C^0(X; R)$
Branch voltage v	1-cochain $v \in C^1(X; R)$
Voltage from a potential	$v = d\phi$
KCL (internal)	$\partial i = 0$ in $C_0(X, T; R)$
KVL (no induction)	v is exact; $[v] = 0$ in $H^1(X; R)$

3.2 Kirchhoff laws and potentials

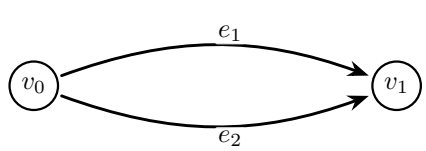
Proof sketch of Theorem 1.1. The chain complex of a graph is $0 \to C_1(X; R) \xrightarrow{\partial} C_0(X; R) \to 0$. Thus $\partial i = 0$ is exactly the statement that currents are 1-cycles (and $\partial i = 0$ in $C_0(X, T; R)$ is the relative version allowing sources at terminals). For voltages, the coboundary $d\colon C^0(X; R) \to C^1(X; R)$ is the transpose incidence map in a chosen basis. If $v = d\phi$, then $v(\gamma) = 0$ on every cycle γ (telescoping sum). Conversely, if v evaluates to 0 on all cycles, then integrating v along a path from a base vertex defines ϕ; the cycle condition ensures path independence, so $v = d\phi$.

3.3 Terminals and "voltage between terminals" as a relative pairing

Fix terminals $T \subset V$. A probe path between two terminals is modeled by a 1-chain γ with $\partial\gamma \in C_0(T; R)$. The voltage readout along that probe is $v(\gamma)$. If $v = d\phi$ then $v(\gamma) = \phi(t_1) - \phi(t_0)$ depends only on the terminal endpoints. More generally, relative homology $H_1(X, T; R)$ packages paths modulo deformations through the interior, and the Kronecker pairing with $H^1(X, T; R)$ is the natural topological home for terminal readouts.

3.4 Worked example: a one-loop graph (explicit matrices)

Consider the "parallel-edge" graph with two vertices v_0, v_1 and two edges e_1, e_2 oriented from v_0 to v_1:



KCL (cycle space). With bases (e_1, e_2) for $C_1 \cong R^2$ and (v_0, v_1) for $C_0 \cong R^2$, the boundary map is

$$\partial = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix},$$

so for a current $i = (i_1, i_2) \in C_1(X; R)$ we have

$$\partial(i_1, i_2) = (-i_1 - i_2)\, v_0 + (i_1 + i_2)\, v_1.$$

If there are no external injections at the nodes (equivalently, $T = \emptyset$ and KCL is $\partial i = 0$), then $i_2 = -i_1$. Thus the cycle space is

$$Z_1(X; R) = \ker \partial = \operatorname{span}\{(1, -1)\},$$

which is rank 1 (one independent loop current).

KVL (exact voltages). The coboundary $d\colon C^0 \to C^1$ is represented by the transpose matrix

$$d = \partial^\top = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix},$$

so $\operatorname{im}(d) = \operatorname{span}\{(1, 1)\} \subset C^1 \cong R^2$. Concretely, writing $\phi = (\phi_0, \phi_1) \in C^0(X; R)$ gives

$$d\phi = (\phi_1 - \phi_0,\ \phi_1 - \phi_0),$$

so a voltage assignment $v = (v_1, v_2)$ is exact (hence satisfies electrostatic KVL) if and only if $v_1 = v_2$.

A cohomology obstruction as "induced EMF". Since $H^1(X; R) \cong C^1/\operatorname{im}(d) \cong R$, there are non-exact 1-cochains. For instance, $v = (1, -1)$ is not exact. Evaluating on the fundamental cycle $\gamma = (1, -1) \in Z_1$ gives the loop integral

$$v(\gamma) = v_1\gamma_1 + v_2\gamma_2 = 1 \cdot 1 + (-1) \cdot (-1) = 2 \neq 0.$$

Engineering interpretation: a nonzero loop integral represents an electromotive force around the loop (e.g. due to time-varying magnetic flux), i.e. precisely the situation in which the electrostatic KVL assumption fails.

4 Field-domain model and measurement

This section introduces the ambient 3-dimensional setup needed to explain energy storage and idealized measurement in genuinely topological terms.

4.1 Conductor regions and the field domain

Let S^3 denote the one-point compactification of $\mathbb{R}^3$.

Definition 4.1 (Conductor region)**.** A *(perfect) conductor region* is a compact 3-dimensional submanifold-with-boundary

$$M \subset S^3$$

whose connected components are the individual conducting bodies (wires, plates, coils, etc.). We write

$$\partial M = \bigsqcup_{\alpha=1}^{m} \Sigma_\alpha$$

for its boundary components.

Definition 4.2 (Field domain)**.** The *field domain* (exterior region) is

$$U := S^3 \setminus \mathrm{int}(M),$$

which is a compact 3-manifold-with-boundary with $\partial U = \partial M$.

Remark 4.1. The S^3-model avoids boundary conditions at spatial infinity; one can instead work in $\mathbb{R}^3$ with decay conditions without changing the purely topological statements below.

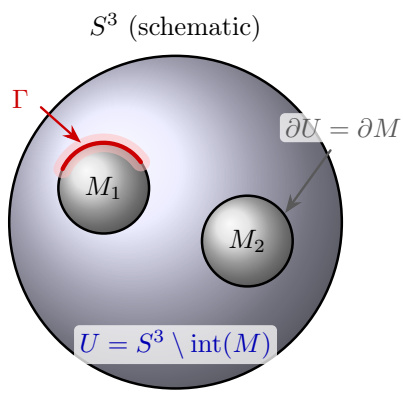


Figure 1: A schematic of the conductor region $M \subset S^3$ and the field domain $U = S^3 \setminus \mathrm{int}(M)$. Terminals are modeled as a distinguished subset $\Gamma \subset \partial U = \partial M$ where boundary conditions and probes are imposed.

4.2 Terminals and relative groups

To model a component, one must specify which boundary components are treated as terminals. Accordingly, we allow a decomposition

$$\partial U = \Gamma \sqcup (\partial U \setminus \Gamma),$$

where Γ is the union of terminal surfaces. The relevant state and measurement spaces are relative groups $H^\bullet(U, \Gamma; R)$ and $H_\bullet(U, \partial U \setminus \Gamma; R)$, paired by Poincaré–Lefschetz duality.

4.3 Measurement as constraint + pairing

Topological content (pairing). Idealized measurement has two parts: a constraint ("do not disturb") and a readout. The readout is topological: it is evaluation of an "effort" class on a "flow" class, as in Theorem 1.4.

Graph level (lumped probes). In the graph model, a voltmeter between terminals can be represented by a relative 1-cycle class in $H_1(X, T)$ (a probe path), and its readout is evaluation of a voltage 1-cochain on that class. An ammeter is represented dually by choosing a cut/edge and evaluating a current 1-chain.

Field level (Poincaré–Lefschetz + Hodge). In the field model, a voltmeter corresponds to evaluating a (relative) electric 1-form class $[E] \in H^1(U, \Gamma)$ on a relative 1-cycle $\gamma \in H_1(U, \partial U \setminus \Gamma)$:

$$\mathrm{Volt}([E], \gamma) = \int_\gamma E.$$

An ammeter corresponds to evaluating a current-flux 2-class on an appropriate spanning surface. Geometry/material data enters through $\star_{\mathrm{mat}}$, which turns these evaluation pairings into inner products and variational principles (minimization of energy or dissipation).

4.4 Poincaré–Lefschetz pairing vs. Hodge star (why measurement needs a metric)

Poincaré–Lefschetz duality provides a *topological* pairing between effort-type and flow-type state spaces:

$$H^k(U, \Gamma; R) \times H_{3-k}(U, \partial U \setminus \Gamma; R) \to R,$$

defined by evaluation/integration on representatives. This pairing is metric-free: it exists as soon as the relevant (co)homology groups are defined.

The Hodge star enters at a different level. Once a metric and material parameters are chosen, the constitutive Hodge operator $\star_{\mathrm{mat}}$ identifies k-forms with $(3-k)$-forms and turns evaluation into an inner product:

$$\langle \alpha, \beta \rangle_{\mathrm{mat}} := \int_U \alpha \wedge (\star_{\mathrm{mat}} \beta).$$

In Hodge theory, harmonic representatives identify cohomology with kernels of Laplacians, and $\star_{\mathrm{mat}}$ maps harmonic k-forms to harmonic $(3-k)$-forms; in this sense it realizes Poincaré–Lefschetz duality as a concrete "Hodge dual" [2, 6]. But unlike Poincaré–Lefschetz duality, $\star_{\mathrm{mat}}$ is not topological: it depends on geometry and material calibration, which is why it is the right home for numerical readout and "least disturbance" measurement principles.

Periods and gauge/constraints. If α is a closed k-form and γ is a k-cycle, then the *period*

$$\int_\gamma \alpha$$

depends only on the cohomology class $[\alpha] \in H^k$ and the homology class $[\gamma] \in H_k$. Periods are therefore the canonical invariants extracted by measurement pairings.

In circuit modeling one often sets certain periods to zero by assumption. For example, electrostatic KVL corresponds to restricting to exact electric fields (vanishing periods around loops), i.e. "no

induced EMF". Similarly, overall potential offsets (the reduced H^0 direction) are fixed by grounding conventions. More generally, "set to zero" enters through boundary conditions at terminals, gauge fixing, or by restricting attention to a subspace of classes appropriate to the regime (electrostatic vs. magnetostatic vs. fully time-dependent).

4.5 From topology to numbers: Hodge and energy

The topology of U determines finite-dimensional state spaces (ranks of the relevant (co)homology groups). Numerical parameters arise after choosing material data, schematically represented by functions/tensors

$$\epsilon\colon U \to \mathbb{R}_{>0}, \qquad \mu\colon U \to \mathbb{R}_{>0}, \qquad \sigma\colon U \to \mathbb{R}_{\geq 0}.$$

These are *not* coefficients of singular cohomology; rather they build bilinear forms by integration. For example, the material Hodge operator $\star_{\mathrm{mat}}$ yields a symmetric form

$$([\alpha],[\beta]) \longmapsto \int_U \alpha \wedge (\star_{\mathrm{mat}}\beta)$$

on the appropriate cohomology group. Choosing bases of topological classes (charges/fluxes/loops) turns these forms into matrices; those matrices are the lumped C and L parameters. Similarly, dissipation $\int_U \sigma|E|^2\,dV$ produces effective resistive operators.

5 Gluing and effective elements

Circuit building is a gluing operation: we form a larger device by joining subdevices along interfaces. On the topological side, the bookkeeping tools are excision and Mayer–Vietoris. On the analytic side, effective "black-box" behavior is encoded by Dirichlet-to-Neumann maps and Schur complements.

5.1 Excision (local modifications do not change relative groups)

Excision says that relative (co)homology is insensitive to removing subsets away from the region where boundary conditions are imposed. Informally, if (X, A) is a pair and $B \subset A$ lies in the interior of A, then the inclusion

$$(X \setminus B,\ A \setminus B) \hookrightarrow (X, A)$$

induces isomorphisms on relative homology and cohomology. In our setting, this justifies replacing complicated conductor shapes by deformation retracts and simplifying U away from terminals, as long as we keep fixed the terminal region Γ.

5.2 Mayer–Vietoris (cut-and-paste exact sequences)

Let $X = A \cup B$ be a union of subspaces with "nice" intersection $A \cap B$. Then Mayer–Vietoris provides a long exact sequence relating $H_\bullet(X)$ to $H_\bullet(A)$, $H_\bullet(B)$, and $H_\bullet(A \cap B)$ (and similarly for cohomology).

Concretely, for homology one has an exact sequence of the form

$$\cdots \longrightarrow H_n(A \cap B) \longrightarrow H_n(A) \oplus H_n(B) \longrightarrow H_n(X) \longrightarrow H_{n-1}(A \cap B) \longrightarrow \cdots,$$

and for cohomology one has

$$\cdots \longrightarrow H^{n-1}(A \cap B) \longrightarrow H^n(X) \longrightarrow H^n(A) \oplus H^n(B) \longrightarrow H^n(A \cap B) \longrightarrow \cdots.$$

Interpretation for circuits. If a device is decomposed into subdevices whose field domains overlap on an interface, Mayer–Vietoris encodes compatibility constraints and identifies how independent degrees of freedom in each piece combine (or are killed) in the union.

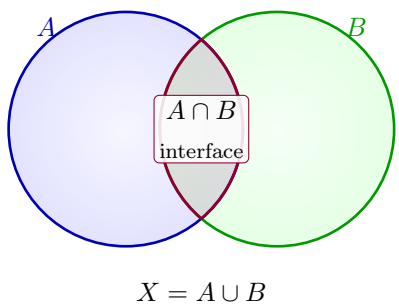


Figure 2: A Mayer–Vietoris decomposition $X = A \cup B$ with overlap $A \cap B$ representing an interface for gluing.

Proposition 5.1 (Direct sums vs. constraints (a Mayer–Vietoris dichotomy))**.** *In favorable decompositions, the relevant part of the Mayer–Vietoris sequence reduces to one of two patterns:*

1. Direct sum. *If the overlap $A \cap B$ is homologically trivial in the degree of interest, then $H_k(X)$ (or $H^k(X)$) is isomorphic to a direct sum of the corresponding groups of A and B.*

2. Matching constraint. *If the overlap carries nontrivial data in the degree of interest, then $H^k(X)$ is naturally identified with a kernel (or quotient) enforcing a matching condition across the interface, e.g. as a kernel of a "restriction-difference" map $H^k(A) \oplus H^k(B) \to H^k(A \cap B)$.*

Proof sketch. Both patterns are immediate from exactness: if $H_\bullet(A \cap B)$ vanishes in the relevant degrees, then the connecting maps in Mayer–Vietoris force the middle map to be an isomorphism (direct sum). If instead the overlap is nontrivial, exactness identifies the global group with the appropriate kernel/quotient enforcing consistency of restrictions on $A \cap B$.

5.3 Dirichlet-to-Neumann maps and Schur complements

Proof idea for Theorem 1.5. Given a quadratic functional in the bulk (energy or dissipation), fixing terminal data and minimizing in the interior produces an induced quadratic form on the terminal space. This is a Dirichlet-to-Neumann (DtN) construction; gluing corresponds to eliminating interface variables, which is a Schur complement.

Resistive DtN (effective conductance/resistance). For dissipation $\mathcal{D}(E) = \int_U \sigma|E|^2\,dV$, minimizing subject to terminal potentials yields a map "terminal voltage $\mapsto$ terminal current"; effective resistance is the inverse of this map on the relevant subspace [1, 10].

Capacitive and inductive DtN (capacitance/inductance matrices). In electrostatics one minimizes $\frac{1}{2}\int_U \epsilon|E|^2\,dV$ subject to boundary potentials on terminal components; the induced terminal charges define the capacitance matrix [1]. In magnetostatics one minimizes the magnetic energy subject to prescribed currents/flux constraints; the induced flux linkages define the inductance matrix [1].

5.4 Series and parallel reductions (topological formulation)

Series/parallel rules are shadows of two basic Mayer–Vietoris behaviors:

- *Direct sums:* when an overlap is homologically trivial in the relevant degree, global state spaces split as direct sums, and quadratic forms add.
- *Constraints/kernels:* when gluing imposes matching conditions across an interface, global state spaces are identified with kernels/quotients, and the effective quadratic form is obtained by minimization (Schur complement).

Interpreting these induced forms in terminal bases yields the familiar additive/reciprocal series/parallel laws.

5.5 Circuit reduction laws via Schur complements

At the lumped level, many textbook reduction laws can be derived from the same variational principle: *add energies of subdevices and eliminate internal variables.* This is exactly the Schur-complement viewpoint of §5.3.

Proposition 5.2 (Series/parallel rules for scalar R, C, L)**.** *Assume ideal, uncoupled elements. Then the effective parameters satisfy:*

- ***Resistors.*** $R_{\text{series}} = R_1 + R_2$ *and* $G_{\text{parallel}} = G_1 + G_2$ *where* $G_i = R_i^{-1}$.
- ***Capacitors.*** $C_{\text{parallel}} = C_1 + C_2$ *and* $C_{\text{series}} = (C_1^{-1} + C_2^{-1})^{-1}$.
- ***Inductors (no mutual coupling).*** $L_{\text{series}} = L_1 + L_2$ *and* $L_{\text{parallel}} = (L_1^{-1} + L_2^{-1})^{-1}$.

Proof sketch. For a two-terminal resistor of conductance G, the dissipation is a quadratic form in the terminal voltage: $P = \frac{1}{2}G(V_a - V_b)^2$. Parallel composition adds dissipation forms directly, giving $G_{\text{parallel}} = G_1 + G_2$. Series composition introduces an internal node potential V_m and one minimizes $P_1(V_a - V_m) + P_2(V_m - V_b)$ over V_m; the resulting induced quadratic form has coefficient $G_{\text{series}} = (G_1^{-1} + G_2^{-1})^{-1}$, i.e. $R_{\text{series}} = R_1 + R_2$.

Capacitors behave identically at the level of energy in voltages: $E = \frac{1}{2}C(V_a - V_b)^2$, so the same elimination produces $C_{\text{parallel}} = C_1 + C_2$ and $C_{\text{series}} = (C_1^{-1} + C_2^{-1})^{-1}$. Inductors are dual: energy is quadratic in currents, $E = \frac{1}{2}LI^2$. Series forces the same current and energies add; parallel splits current and one minimizes $\frac{1}{2}L_1I_1^2 + \frac{1}{2}L_2I_2^2$ subject to $I = I_1 + I_2$, yielding $L_{\text{parallel}} = (L_1^{-1} + L_2^{-1})^{-1}$.

Matrix remark. For multi-terminal linear passive devices, the same statements hold with scalars replaced by symmetric positive (semi)definite terminal operators and with series/parallel realized as block sums plus Schur complements (the DtN viewpoint).

5.6 Δ/Y (star–triangle) as Schur complement

Delta–wye equivalence is a canonical example of "odd gluing": a 3-terminal network is simplified by eliminating an internal node. Historically this is the star–triangle transform [11]; mathematically it is a Schur complement of a discrete DtN/response operator [12].

Operator viewpoint (eliminate the internal node). Consider a star (wye) network with terminals a, b, c connected to an internal node 0 by conductances g_a, g_b, g_c. Writing node potentials as (V_0, V_a, V_b, V_c), the graph Laplacian (nodal admittance matrix) has the block form

$$L = \begin{pmatrix} g_a + g_b + g_c & -g_a & -g_b & -g_c \\ -g_a & g_a & 0 & 0 \\ -g_b & 0 & g_b & 0 \\ -g_c & 0 & 0 & g_c \end{pmatrix}.$$

Eliminating V_0 (Schur complement) produces an effective 3×3 terminal operator whose off-diagonal entries are $-g_{ab}, -g_{bc}, -g_{ca}$, i.e. an equivalent triangle (delta) network.

Classical conductance formulas. Let $g_\Sigma := g_a + g_b + g_c$. The equivalent triangle conductances are

$$g_{ab} = \frac{g_a g_b}{g_\Sigma}, \qquad g_{bc} = \frac{g_b g_c}{g_\Sigma}, \qquad g_{ca} = \frac{g_c g_a}{g_\Sigma}.$$

(Equivalently, in resistances $R_i = g_i^{-1}$ one obtains $R_{ab} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_c}$ and cyclic permutations.)

Remark (beyond resistors). The same Schur-complement algebra applies to any linear passive terminal operator (capacitance matrices, impedance/admittance matrices at fixed frequency), but the resistive case is the cleanest place to see the topology/variational structure explicitly.

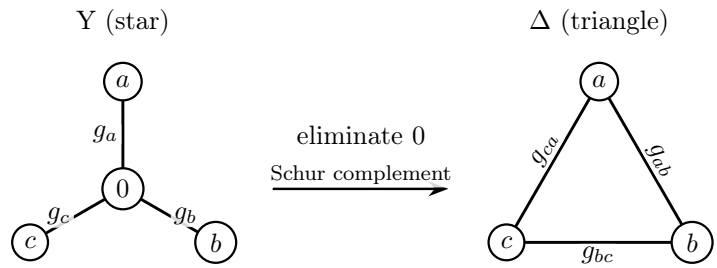


Figure 3: Delta–wye (star–triangle) equivalence as elimination of an internal node (Schur complement of the response/Laplacian matrix).

5.7 Optional: a categorical viewpoint

Many network results become slick once one treats devices not as collections of equations but as maps between boundary data spaces. In this paper those boundary data spaces are (finite-dimensional) (co)homology groups associated to terminals, and the relevant maps are induced by restriction, duality pairings, and Hodge/constitutive operators. Gluing corresponds to composition; exactness of Mayer–Vietoris is the mechanism ensuring the composition is consistent.

6 Alexander duality and degrees of freedom for C and L

Alexander duality is the basic bridge between the topology of the conductor region $M \subset S^3$ and the topology of the field domain $U = S^3 \setminus \mathrm{int}(M)$. It is the mechanism by which "capacitor-type" disconnectedness of M and "inductor-type" linking/loop phenomena in U become the same statement in different degrees.

Proof sketch of Theorem 1.3. Alexander duality identifies reduced cohomology of the complement with reduced homology of the subset in complementary degree. Disconnectedness of M is detected by $\widetilde{H}_0(M)$ and produces degree-2 classes in U (capacitive degrees of freedom), while loop classes in M (and tunnels in the complement) produce degree-1 classes in U (inductive degrees of freedom). Once ϵ and μ are chosen, these state spaces acquire bilinear forms by integration (Section 4.5); effective C/L matrices are obtained by DtN reduction (Section 5.3).

6.1 Statement (informal form)

Let $A \subset S^3$ be a compact, locally contractible subset (in particular, a compact submanifold-with-boundary as in our conductor model). Then Alexander duality provides natural isomorphisms for reduced (co)homology:

$$\widetilde{H}^k(S^3 \setminus A;\, R) \;\cong\; \widetilde{H}_{2-k}(A;\, R),$$

for all k, where R is a coefficient ring/field. In our setting, $A = M$ (or sometimes a deformation retract of M capturing its topology).

6.2 Two circuit-relevant consequences

Capacitors (components of M). If M has multiple connected components, then $\widetilde{H}_0(M;R) \neq 0$, hence by Alexander duality $\widetilde{H}^2(S^3 \setminus M;R) \neq 0$. This is the topological reason that separating 2-classes in the field domain exist when there are multiple conductors: they support nontrivial electric-flux degrees of freedom (which become a capacitance matrix once ϵ is chosen).

Inductors (loops/linking in the complement). Nontrivial $\widetilde{H}_1(S^3 \setminus M;R)$ (equivalently $H_1(U;R)$) corresponds dually to $\widetilde{H}^1(S^3 \setminus M;R) \cong \widetilde{H}_1(M;R)$ in the simplest cases, and more generally is detected by the presence of nontrivial "tunnels" in the complement. This is the topological home for magnetic flux/linking effects that, after choosing μ, become the inductance matrix.

Figure 4: Cartoons of the topological degrees of freedom behind a capacitor (left: a separating surface class in $H^2(U)$ associated to disconnected conductors) and an inductor (right: linking detected by intersection of a cycle with a spanning surface).

6.3 Relative refinements and terminals

For actual circuit elements we typically distinguish terminal surfaces $\Gamma \subset \partial M$ and impose boundary conditions there. The correct statements then involve relative (co)homology and Poincaré–Lefschetz duality on U (a manifold with boundary), together with excision to pass between $S^3 \setminus M$ and U. We will use these relative refinements when deriving capacitance/inductance matrices for multi-terminal devices.

6.4 Alexander duality as a "degree-of-freedom counter" (and when reductions exist)

Alexander duality is most useful here as a clean *counting principle*: it predicts which finite-dimensional topological state spaces *can* exist before any constitutive geometry is chosen. In particular, it tells you which "reactive" degrees of freedom are even available to be turned into C and L matrices by a Hodge/energy construction.

Why this helps with reductions. Interconnection operations (series/parallel attachments, identification of terminal components, adding/removing internal nodes) change the topology of the con-

ductor region M in controlled ways, hence change $\widetilde{H}_0(M)$ and $\widetilde{H}_1(M)$. By Alexander duality, this forces corresponding changes in the complement groups (schematically, $H^2(U)$ for capacitive classes and $H^1(U)$ for inductive classes). Thus topology predicts *which* reductions are meaningful: one can only reduce if the gluing keeps track of the relevant state spaces and their pairings.

For the actual reduction *formulas* (series/parallel and Δ/Y) one needs both Mayer–Vietoris bookkeeping and the variational/DtN mechanism: see Section 5, especially §5.4 and §5.6.

7 Time topology and AC/DC signals

We model time dependence by taking a product with a 1-manifold T:

- *DC* (transients/step response): $T = [0, \infty]$ (homeomorphic to an interval) with boundary.
- *AC* (steady periodic forcing): $T = S^1$ with $\partial T = \emptyset$.

The relevant (co)homology of T is elementary:

$$H_0([0,\infty]; R) \cong R, \qquad H_1([0,\infty]; R) = 0,$$

and

$$H_0(S^1; R) \cong R, \qquad H_1(S^1; R) \cong R.$$

For any space Y used to model spatial/topological degrees of freedom (e.g. the circuit complex X or a field-domain pair (U, Γ)), the product $Y \times T$ organizes time-dependent degrees of freedom.

7.1 Topology-only constraints for DC vs. AC (what changes when time is S^1)

Proof sketch of Theorem 1.2. Over a field, the Künneth theorem gives a direct-sum decomposition

$$H_k(Y \times T; R) \cong \bigoplus_{i+j=k} H_i(Y; R) \otimes H_j(T; R),$$

and similarly for cohomology. Passing from DC (T contractible) to AC ($T = S^1$) adds an $H_1(S^1)$ factor coming from the time circle.

Even before choosing a Fourier basis, periodic time enlarges natural state spaces. For example, if $H^k(Y; R) \neq 0$, then Künneth produces additional cohomology in degree $k+1$:

$$H^{k+1}(Y \times S^1; R) \ \supset \ H^k(Y; R) \otimes H^1(S^1; R).$$

Engineering intuition: why the $H^1(S^1)$ factor matters. In the de Rham picture, a generator of $H^1(S^1; R)$ is represented by the closed 1-form dt (normalized so $\int_{S^1} dt = 1$). Thus any spatial class $[\alpha] \in H^k(Y; R)$ gives a degree-shifted class

$$[\alpha \wedge dt] \in H^{k+1}(Y \times S^1; R),$$

which is precisely the $H^k(Y) \otimes H^1(S^1)$ summand above. Conceptually, periodic forcing introduces an additional "phase direction": spatial degrees of freedom acquire a canonical partner involving the time-circle direction, which is the topological shadow of the in-phase/quadrature splitting that engineers package as complex phasors. The slant product with $[S^1]$ (Section 2) is the corresponding "integrate over one period" operation: it extracts time-averaged/period-integral quantities from classes on $Y \times S^1$.

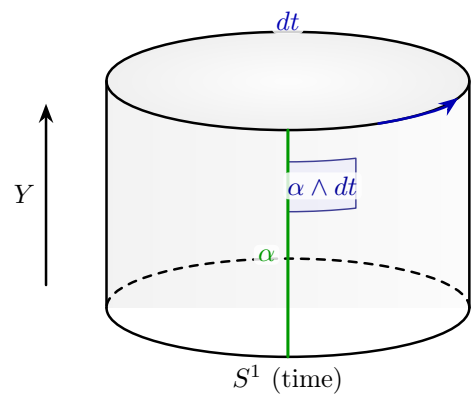


Figure 5: Cartoon of the Künneth summand $H^k(Y) \otimes H^1(S^1)$: a spatial class α and the time-circle generator dt combine to produce a degree-shifted class $\alpha \wedge dt$ on $Y \times S^1$.

Homology of S^1 versus Fourier modes. Singular homology sees only the constant class and the winding class. The integer-indexed Fourier modes arise analytically as characters of the circle group (or eigenfunctions of the Laplacian on S^1), not as additional homology classes. The topological role of S^1 is to provide a new degree-1 time direction that can be tensored with spatial classes; the analytic role is that differentiation on each Fourier mode inserts an $i\omega$ factor.

7.2 Leading/lagging and phase as "90° rotation" on the time circle

The most visible dynamical signature of energy storage is phase shift: in sinusoidal steady state, ideal capacitors make current lead voltage, while ideal inductors make current lag voltage.

Time derivatives and S^1-modes. Working on $T = S^1$ with coordinate t, the operator $\frac{d}{dt}$ acts on Fourier modes e^{int} by multiplication by in. Thus, mode-by-mode, differentiation corresponds to a 90° phase shift.

Ideal capacitor and inductor (mode-by-mode). The constitutive laws are

$$i_C = C\,\frac{dv_C}{dt}, \qquad v_L = L\,\frac{di_L}{dt}.$$

On a single Fourier mode of angular frequency ω, these become

$$i_C = (i\omega C)\,v_C, \qquad v_L = (i\omega L)\,i_L,$$

so $Z_C(\omega) = \frac{1}{i\omega C}$ and $Z_L(\omega) = i\omega L$.

What is (and is not) topological about the 90° shift. The existence of a capacitive or inductive *state variable* is topological (it is detected by the appropriate (co)homology group being nontrivial). However, the quadrature phase shift itself does not come from (co)homology alone: it comes from universal interconnection laws introducing a time derivative,

$$i = \frac{dq}{dt}, \qquad v = \frac{d\Phi}{dt},$$

and hence an $i\omega$ in the frequency domain. In this sense topology tells us *that* there is a storage coordinate (q or Φ), while the time circle tells us *how* it rotates effort/flow variables by 90° on each mode.

Hodge star as a spatial duality (separate from the time rotation). The Hodge/constitutive operator $\star_{\mathrm{mat}}$ provides an additional duality on spatial forms and induces positive definite bilinear forms on cohomology by

$$([\alpha], [\beta]) \longmapsto \int_U \alpha \wedge (\star_{\mathrm{mat}}\beta).$$

This is the mechanism that turns a purely topological degree of freedom into a *numerical* capacitance or inductance, while the 90° phase shift comes from the complex structure on S^1-modes (the $i\omega$ factor).

8 Worked examples and optional extensions

8.1 Worked topology examples

8.1.1 Spherical cavity (absolute/relative homology)

Let M be the spherical shell obtained by removing the unit ball from the ball of radius 2. Then ∂M has two components, each a 2-sphere, and the long exact sequence of the pair $(M, \partial M)$ computes the nontrivial groups:

- $H_3(M, \partial M; R) \cong R$ generated by the fundamental class $[M]$.
- $H_2(\partial M; R) \cong R^2$ generated by the two boundary spheres; likewise $H_0(\partial M; R) \cong R^2$.
- $H_2(M; R) \cong R$ represented by a sphere separating the two boundary components.
- $H_1(M, \partial M; R) \cong R$ represented by a curve connecting the two boundary components.

This is the basic template behind "capacitor" behavior: disconnected terminal components and a separating 2-class paired with a relative 1-class.

8.1.2 Linking, Gauss integrals, and mutual inductance (topological viewpoint)

The simplest genuinely 3-dimensional phenomenon behind inductors is linking. Topologically, linking is detected by homology/cohomology of the complement and by Alexander duality; analytically, it is computed by the Gauss linking integral.

Linking number. Let $\gamma_1, \gamma_2 \subset S^3$ be two disjoint, oriented smooth embedded circles. Their *linking number* $\mathrm{Lk}(\gamma_1, \gamma_2) \in \mathbb{Z}$ can be defined by choosing an oriented spanning surface S with $\partial S = \gamma_1$ and setting

$$\mathrm{Lk}(\gamma_1, \gamma_2) := \#(S \cap \gamma_2)$$

as the signed intersection number.

Gauss linking integral. If γ_1, γ_2 are parametrized in $\mathbb{R}^3$, the same integer is given by

$$\mathrm{Lk}(\gamma_1, \gamma_2) = \frac{1}{4\pi} \int_{\gamma_1} \int_{\gamma_2} \frac{(x-y) \cdot (dx \times dy)}{\|x-y\|^3}.$$

This formula is a concrete instance of a cohomological pairing computing an intersection/linking invariant.

Mutual inductance as a geometrically weighted pairing. In magnetostatics, the flux through a loop induced by current in another loop can be viewed as a symmetric bilinear form on the relevant "loop" state space (typically $H_1(U; R)$ or a relative variant), obtained by composing a topological pairing (linking/intersection, controlled by the complement topology) with a constitutive operator determined by μ (and the metric). Topology controls *which* couplings can exist robustly, while geometry/materials determine their numerical values.

8.2 Worked lumped-circuit examples

These short examples connect the chain/cochain picture to standard computations.

8.2.1 Single-loop current as a rank-one cycle space

Let X be any connected graph with one independent loop (first Betti number 1). Then $\ker \partial \subset C_1(X; R)$ is one-dimensional, so any KCL-satisfying current is a scalar multiple of a fundamental cycle current. Constitutive data (resistive/inductive) selects the scalar via an energy/dissipation minimization or via the time-domain ODE for the storage coordinate.

8.2.2 Two-terminal "voltage" as a relative evaluation

Let $T = \{t_0, t_1\} \subset V$ be two terminals. Any probe path is a 1-chain γ with $\partial\gamma = t_1 - t_0$. If $v = d\phi$ then $v(\gamma) = \phi(t_1) - \phi(t_0)$, independent of the chosen path. When $[v] \neq 0$ in $H^1(X; R)$, different probe paths differ by a cycle and the discrepancy is detected by evaluating $[v]$ on that cycle.

8.2.3 Series/parallel as elimination (Schur complement)

In a linear resistive network, the power functional is a quadratic form in branch currents/voltages. Gluing two subnetworks and eliminating internal node potentials is a Schur complement computation on that quadratic form; the resulting induced form on terminal data is the effective resistance

operator. This is the algebraic shadow of the Mayer–Vietoris "direct sum vs. constraint" dichotomy discussed in §5.4.

8.3 Optional: almost topological degrees of freedom

Ideal topological degrees of freedom are defined by exact closedness/exactness and perfect boundary conditions. Real devices are non-ideal: conductors have finite conductivity, dielectrics leak, and gluing decompositions are only approximately valid. This optional module records one way to make the ideal picture quantitative.

Functional-analytic setting (minimal). In a Maxwell/FEEC-type framework, one works in Sobolev spaces such as $H(\mathrm{curl}, U)$ and $H(\mathrm{div}, U)$ with boundary conditions encoding terminals [6]. The material inner product $\langle \cdot, \cdot \rangle_{\mathrm{mat}}$ defines a (weighted) Hodge Laplacian Δ_{mat} whose kernel corresponds (in ideal lossless settings) to the relevant cohomology classes.

Spectral viewpoint: "almost" as a near-kernel. Non-idealities perturb exact kernels into small but nonzero eigenvalues. Thus one may call a mode "ε-topological" if its Rayleigh quotient

$$\lambda(\alpha) := \frac{\langle \Delta_{\mathrm{mat}} \alpha, \alpha \rangle_{\mathrm{mat}}}{\langle \alpha, \alpha \rangle_{\mathrm{mat}}}$$

is $\leq \varepsilon$. Small eigenvalues correspond to slowly decaying modes (large time constants, high-Q behavior).

Toy example: an ideal capacitor with leakage. An ideal capacitor has a conserved charge coordinate q (a "zero mode"). Adding a small leakage resistance R in parallel turns conservation into decay:

$$\frac{dq}{dt} + \frac{1}{RC}\, q = 0.$$

The eigenvalue 0 is perturbed to $1/(RC)$, illustrating how an "exact" storage mode becomes an "almost" mode with a long decay time.

Further directions (brief). Possible extensions include sheaf-theoretic models for localized nonlinear components, persistent-homology bookkeeping for switching networks, and more explicit functorial links between Mayer–Vietoris gluing and operator Schur complements.

8.4 Optional: higher-order linking and Massey products

Cup and cap products (and their pairings) capture many of the invariants that appear naturally in linear circuit theory: evaluation of a cohomology class on a cycle, intersection/linking numbers, and bilinear energy forms. However, 3-dimensional topology contains genuine *higher* operations that are invisible to pairwise linking.

Borromean rings. The Borromean rings are a 3-component link $L = L_1 \sqcup L_2 \sqcup L_3 \subset S^3$ with the property that each two-component sublink is an unlink (all pairwise linking numbers vanish), but the full three-component link is not trivial. One algebraic-topology way to express the remaining information is via a triple Massey product in the complement $X := S^3 \setminus L$: the meridian classes $a_i \in H^1(X;\mathbb{Z})$ have vanishing pairwise cup products, so the triple product $\langle a_1, a_2, a_3 \rangle$ is defined and is nontrivial. This is closely related to Milnor's $\bar{\mu}$-invariants for links [13, 14].

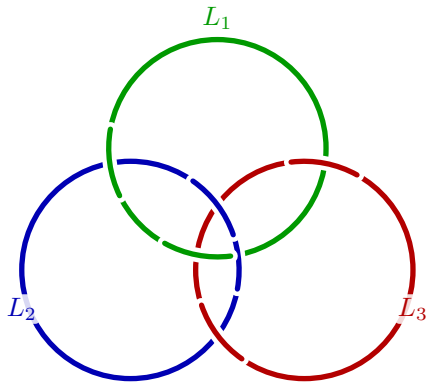


Figure 6: A schematic knot-diagram projection of the Borromean rings. Any two components are unlinked (pairwise linking numbers 0), but there is a nontrivial higher-order linking detected by a triple Massey product in the complement.

Circuit-facing remark (no overclaim). Our linear RLC reduction laws are extracted from quadratic energy/dissipation functionals and their induced terminal operators (DtN maps and Schur complements). Those constructions naturally see bilinear pairings and low-degree (co)homology operations, but they do not by themselves generate new *linear* element laws from higher-order operations such as Massey products. Nevertheless, the Borromean example emphasizes that ambient topology of multi-loop embeddings contains structure beyond pairwise linking, which can become relevant when one goes beyond linear passive elements (e.g. nonlinear materials, multiphysics coupling, or constrained measurement protocols).

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